

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary Examination – Summer 2024 Course: B. Tech. (Common to all Branches) Semester : I Subject Name & Code: Engineering Mathematics – I (BTBS101) Max Marks: 60 Date: 29.06.2024 Duration: 3 Hrs.			
Instructions to the Students: 1. All the questions are compulsory. 2. Use of non-programmable scientific calculators is allowed. 3. Assume suitable data wherever necessary and mention it clearly.			
			Marks
Q. 1	Solve Any Two of the following.		12
A)	Solve the equations: $x + 2y + 4z + w = 0 ; 2x + 4y + 8z + 2w = 0 ; 3x + 6y + 2z = 0$		6
B)	Verify the Cayley – Hamilton theorem for the matrix $A = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$ and hence find A^{-1} . Also determine A^{-1} .		6
C)	Find the eigen values and corresponding eigen vectors of the matrix $A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$.		6
Q.2	Solve Any Two of the following:		12
A)	If $u = \log(x^3 + y^3 + z^3 - 3xyz)$, show that $\left(\frac{\partial}{\partial x} + \frac{\partial}{\partial y} + \frac{\partial}{\partial z}\right)^2 u = \frac{-9}{(x+y+z)^2}$.		6
B)	If z is a homogeneous function of degree n in x and y , then prove that $x^2 \frac{\partial^2 z}{\partial x^2} + 2xy \frac{\partial^2 z}{\partial x \partial y} + y^2 \frac{\partial^2 z}{\partial y^2} = n(n-1)z$.		6
C)	If $u = f(y-z, z-x, x-y)$, show that $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 0$.		6
Q. 3	Solve any Two of the following:		12
A)	If $u = \frac{yz}{x}, v = \frac{zx}{y}, w = \frac{xy}{z}$, show that $\frac{\partial(u,v,w)}{\partial(x,y,z)} = 4$.		6
B)	Expand $f(x, y) = e^{xy}$ at $(1, 1)$ by using Taylor's theorem.		6
C)	Find the points on the surface $z^2 = xy + 1$ nearest to the origin.		6
Q.4	Solve any Two of the following:		12
A)	Evaluate $\int_0^{\frac{\pi}{2}} \sin^8 x \cos^7 x \, dx$.		6
B)	Trace the curve $y^2 = x^3$		6
C)	Trace the curve $r\theta = a$.		6
Q. 5	Solve any Two of the following:		12
A)	Evaluate the integral $I = \int_1^a \int_1^b \frac{dy \, dx}{xy}$		6
B)	Change the order of integration and evaluate $\int_0^{\frac{\pi}{2}} \int_x^{\frac{\pi}{2}} \left(\frac{\cos y}{y}\right) \, dx \, dy$.		6
C)	Evaluate the integral $I = \int_0^a \int_0^x \int_0^{x+y} e^{x+y+z} \, dz \, dx \, dy$		6
*** End ***			